



CSM SYNERGY

# BuildBridge

## GoHighLevel Marketplace App Platform

A multi-tenant integration platform that connects GoHighLevel sub-accounts to building-industry tools — brokering authentication, subscriptions, field mapping, and live data sync between GHL, Deposyt billing, and SmartBuild.

Node.js + Express 5

React 19 + Vite

PostgreSQL + Drizzle

Railway / Docker

Technical Overview • Repository: [CSMSYNERGY/buildbridge](#) • Generated 29 Jun 2026

# 1. What BuildBridge Is

**BuildBridge** (internal package name `smartbuild-v2` ) is a **GoHighLevel (GHL) marketplace application platform** built by CSM Synergy. It is a multi-tenant SaaS: GHL agencies install it from the marketplace, and each GHL *location* (sub-account) can then subscribe to integration "apps," configure them, and have BuildBridge act as the middleware that moves data between GHL and third-party building-industry software — primarily **SmartBuild**, a building/job estimation system.

### The Problem It Solves

Construction & outdoor-structure businesses run their sales/CRM in GoHighLevel but their quoting/estimating in separate tools like SmartBuild. BuildBridge bridges the two so a job created or updated in one is reflected in the other — without manual re-entry.

### How It Earns Money

Per-location subscriptions billed through **Deposyt**. Each app (SmartBuild, IdeaRoom, QuickBooks, Monday.com) has monthly/annual plans, plus a discounted **Suite** plan that unlocks everything. Access is gated by subscription status.

## Repository Snapshot

| Attribute         | Value                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Repository        | github.com/CSMSYNERGY/buildbridge (default branch <code>main</code> )                                                              |
| Package / version | <code>smartbuild-v2</code> v2.0.0                                                                                                  |
| Structure         | Monorepo — Express backend ( <code>src/</code> ) + React SPA ( <code>frontend/</code> )                                            |
| Runtime           | Node.js ≥ 20, ES Modules                                                                                                           |
| Deployment        | Railway via multi-stage <code>Dockerfile</code> ; health check at <code>/health</code>                                             |
| Database          | PostgreSQL (Supabase) via Drizzle ORM                                                                                              |
| Maturity          | Actively developed, early stage — core auth/billing/config flows complete; a few action endpoints still stubbed; no test suite yet |

## Technology Stack

### Backend

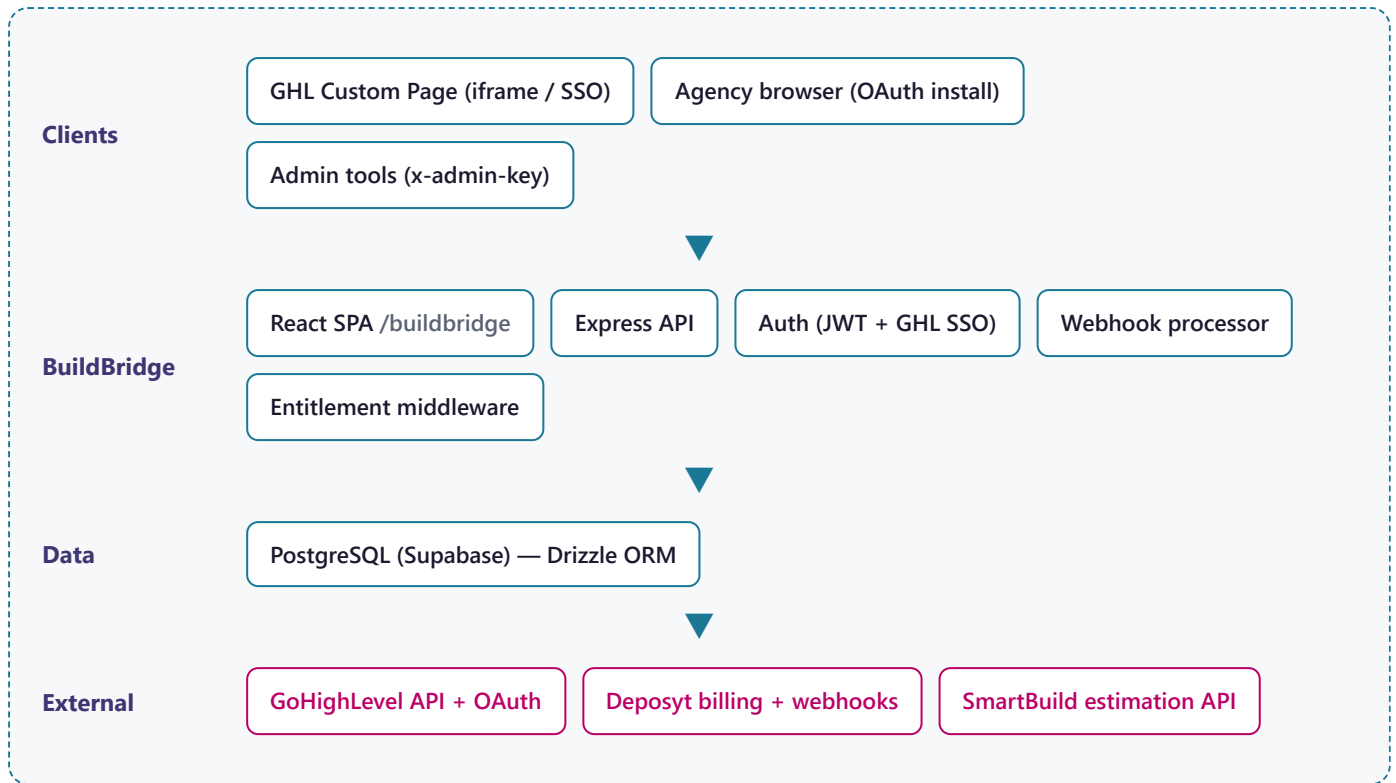
- Express 5 REST API
- Drizzle ORM + `postgres-js`
- JWT sessions ( `jsonwebtoken` , httpOnly cookie)
- Helmet, CORS, rate limiting, morgan logging
- AES-256-GCM encryption for secrets at rest
- `invalid` for env-var validation

### Frontend

- React 19 + React Router 7
- Vite 6 build, served at `/buildbridge`
- Tailwind CSS + Radix UI primitives
- Lucide-react icons, cookie-based auth context
- Brand: purple `#3d3672` / teal `#75e6da`

## 2. Architecture

BuildBridge is a single Node service that serves both the REST API and the compiled React SPA. It sits between GoHighLevel (identity + CRM data), Deposyt (billing), and SmartBuild (estimation), persisting state in PostgreSQL.



### End-to-End Flows

- 1 Install & OAuth.** An agency installs from the GHL marketplace. `GET /auth` redirects to GHL's OAuth chooser; `GET /auth/callback` exchanges the code for access/refresh tokens, encrypts them (AES-256-GCM), upserts the `locations` row, and issues a JWT session cookie ( `sb_token` ).
- 2 In-app SSO.** Inside GHL, the app loads as a Custom Page in an iframe. `/api/sso/decrypt` decrypts GHL's SSO key (AES-256-CBC w/ shared secret) to identify the user per request and set the same session cookie. Helmet CSP/frameguard are relaxed to permit iframe embedding.
- 3 Billing.** Subscriptions are created/canceled via the Deposyt API. Deposyt fires HMAC-SHA256–signed webhooks to `/webhooks/subscription`, which are signature-verified, idempotency-checked, logged to `webhook_events`, then processed (add / update / delete / pause).
- 4 Entitlement.** The `checkSubscription(appSlug)` middleware gates features by active plan. A **Suite** subscription unlocks all apps (smartbuild, idearoom, quickbooks, monday).
- 5 Integration actions.** `/actions/*` (API-key + JWT + subscription) proxy into SmartBuild — retrieve / create / edit a job. SmartBuild credentials are stored encrypted per-location and exchanged for a session token on each call.

### Security Model

- Secrets at rest:** GHL tokens and SmartBuild credentials encrypted with AES-256-GCM before DB storage.
- Sessions:** signed JWT in an httpOnly, SameSite=None, Secure cookie (7-day expiry).
- Webhooks:** HMAC signature verified with `timingSafeEqual`; replay-protected via an idempotency event log.
- Internal/admin routes:** guarded by `X-API-Key` / `x-admin-key` headers.
- Hardening:** Helmet headers, CORS locked to the app origin, and per-route rate limiters.

## 3. Data Model & API

### Database Schema (Drizzle / PostgreSQL)

| Table                   | Purpose                                   | Key columns                                                                                                                                                                                                                                          |
|-------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| plans                   | Sellable subscription plans per app       | id, appSlug, billingInterval, priceUsd (cents), isActive                                                                                                                                                                                             |
| locations               | GHL sub-accounts + encrypted OAuth tokens | id (GHL locationId), companyId, ghlAccessToken  , ghlRefreshToken  , expiresAt |
| subscriptions           | Active/cancelled plans per location       | id (Deposyt sub id), locationId→, planId→, status, period start/end                                                                                                                                                                                  |
| webhook_events          | Idempotent inbound event log              | id (idempotency key), source, eventType, payload(jsonb), status                                                                                                                                                                                      |
| mappers                 | Field mapping GHL ↔ external app          | locationId→, appSlug, mapperType, externalKey, ghlValue (unique)                                                                                                                                                                                     |
| integration_credentials | Encrypted per-app secrets                 | locationId→, appSlug, encryptedPayload  (unique per location+app)                                                                                                 |

 = encrypted (AES-256-GCM) • → = foreign key to **locations** / **plans** . Managed with Drizzle Kit migrations; **npm run seed** loads 10 starter plans (5 apps × monthly/annual + Suite).

### API Surface

| Method & Path                                                              | Auth                | Description                               |
|----------------------------------------------------------------------------|---------------------|-------------------------------------------|
| <b>GET</b> /health                                                         | none                | Liveness probe                            |
| <b>GET</b> /auth · /auth/callback                                          | none                | GHL OAuth start & token-exchange callback |
| <b>GET</b> / <b>POST</b> /api/sso/decrypt                                  | SSO key             | GHL SSO → issue session cookie            |
| <b>GET</b> /api/me                                                         | JWT                 | Current user/location from session        |
| <b>GET</b> /api/subscription/plans                                         | JWT                 | Active plans (raw + grouped by app)       |
| <b>POST</b> /api/subscription/create                                       | JWT                 | Create Deposyt subscription               |
| <b>DEL</b> /api/subscription/cancel                                        | JWT                 | Cancel subscription                       |
| <b>GET</b> /api/ghl/fields                                                 | JWT                 | List GHL custom fields for mapping        |
| <b>GET</b> /api/mappers · CRUD                                             | JWT                 | Create/read/update/delete field mappers   |
| <b>GET</b> /api/smartbuild/config · <b>POST</b> · <b>DEL</b> · <b>test</b> | JWT                 | Manage & test SmartBuild credentials      |
| <b>POST</b> /actions/*                                                     | API-key + JWT + sub | Retrieve / create / edit SmartBuild jobs  |
| <b>POST</b> /webhooks/subscription                                         | HMAC sig            | Deposyt billing webhooks                  |

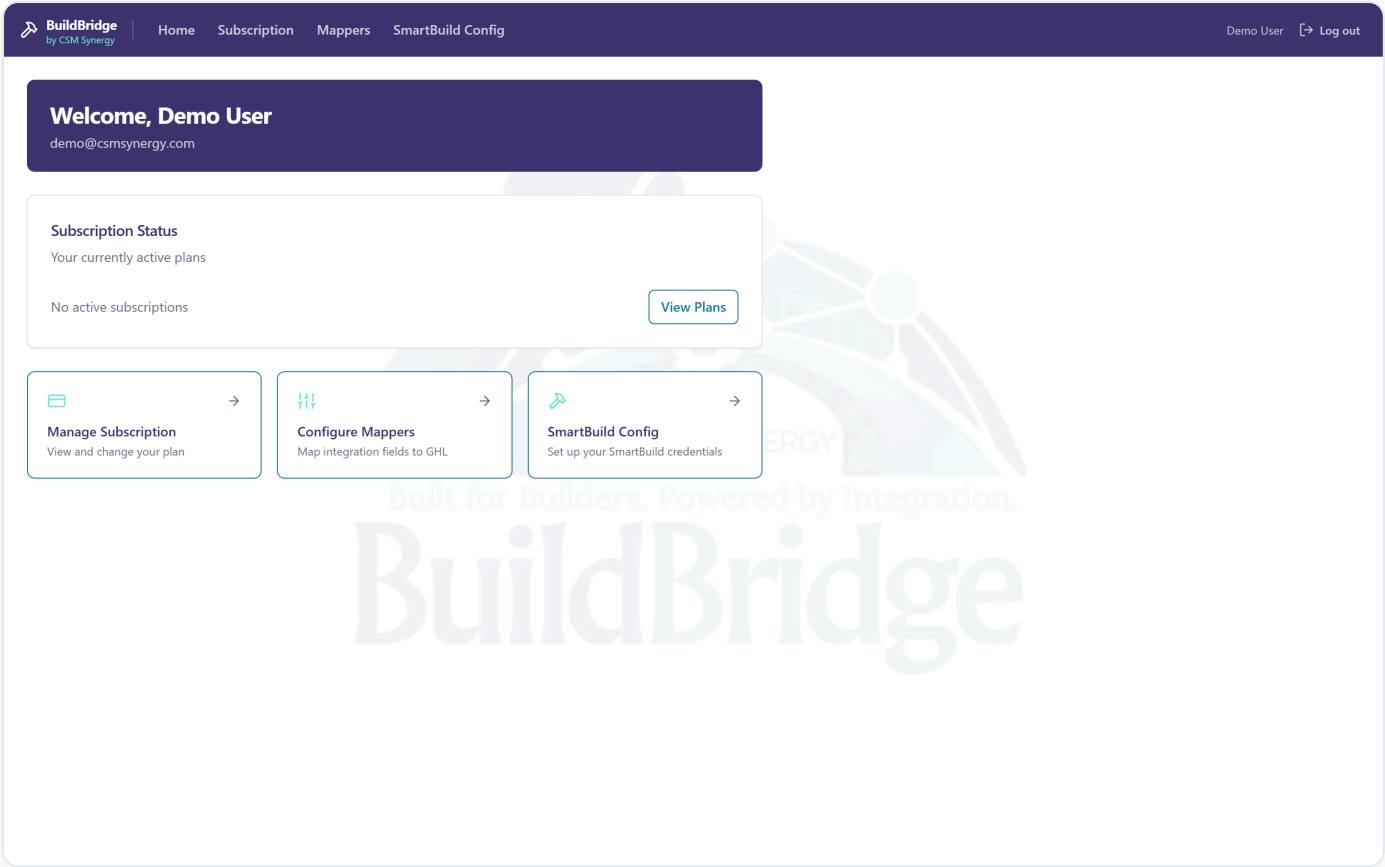
Plan Catalogue (seed data)

| App              | Monthly | Annual  |
|------------------|---------|---------|
| SmartBuild       | \$150   | \$1,500 |
| IdeaRoom         | \$49    | \$490   |
| QuickBooks       | \$69    | \$690   |
| Monday.com       | \$59    | \$590   |
| Suite (all apps) | \$250   | \$2,500 |

## 4. The Application (Screenshots)

Captured from the app running locally. The interface is a clean, branded admin console served at `/buildbridge` with a four-section top navigation.

### Home / Dashboard



Welcome banner, live subscription-status card, and quick-link cards to the three configuration areas. A subtle BuildBridge watermark sits behind the content.

### SmartBuild Config

## SmartBuild Config

Connect your SmartBuild account credentials.

⊗ Not connected

No credentials

### Login Credentials

These credentials are encrypted at rest and never exposed in plaintext. ^

#### Username

your@email.com

#### Password

\*\*\*\*\*

#### SmartBuild Base URL

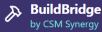
https://app.smartbuild.io



Save Configuration

Connect a SmartBuild account (username / password / base URL). Credentials are encrypted at rest; a connection-status indicator and a "Test connection" action validate them before saving.

## Subscription

 BuildBridge  
by CSM Synergy

[Home](#) [Subscription](#) [Mappers](#) [SmartBuild Config](#)

Demo User [Log out](#)

### Subscription

Choose the plan that fits your workflow.

Monthly

☐ Annual

Save ~17%

SmartBuild

\$150<sub>/mo</sub>

✓ Full SmartBuild integration

✓ GHL workflow actions

Subscribe

IdeaRoom

\$49<sub>/mo</sub>

✓ Full IdeaRoom integration

✓ GHL workflow actions

Subscribe

QuickBooks

\$69<sub>/mo</sub>

✓ Full QuickBooks integration

✓ GHL workflow actions

Subscribe

Monday.com

\$59<sub>/mo</sub>

✓ Full Monday.com integration

✓ GHL workflow actions

Subscribe

Suite (All Apps)

\$250<sub>/mo</sub>

✓ Full Suite (All Apps) integration

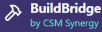
✓ GHL workflow actions

✓ All apps included

Subscribe

Plan selection with a monthly/annual toggle ("Save ~17%" on annual) and per-app pricing cards — shown here populated with live data from the seeded database (5 apps: SmartBuild, IdeaRoom, QuickBooks, Monday.com, Suite).

## Mappers

 BuildBridge  
by CSM Synergy

[Home](#) [Subscription](#) [Mappers](#) [SmartBuild Config](#)

Demo User [Log out](#)

### Mappers

Map integration fields to GoHighLevel values.

+ New Mapper

No mappers configured yet.

Create your first mapper

*CRUD interface for mapping integration fields to GoHighLevel values, shown signed-in with a clean empty state prompting creation of the first mapper.*

## 5. Running It Locally

The repository is cloned at `D:\CSM Synergy\buildbridge` . Both tiers run with Node 20+.

### Backend (Express API)

- `npm install` in the repo root
- Copy `.env.example` → `.env` and fill values (a valid 64-hex `ENCRYPTION_KEY` is required)
- `npm start` (loads dotenv via `src/index.js` )

**Note:** the Docker entrypoint `src/start.js` checks env vars *before* dotenv loads — that's for Railway, which injects env directly. For local dev, use `npm start` , not `node src/start.js` .

### Frontend (Vite SPA)

- `cd frontend && npm install`
- `npm run dev` → serves `http://localhost:5173/buildbridge/` and proxies `/api` , `/auth` , `/webhooks` to the backend

### Database & integrations

- Point `DATABASE_URL` at a PostgreSQL instance, then `npm run migrate` and `npm run seed`
- OAuth, billing, and SmartBuild flows require real GHL / Deposyt / SmartBuild credentials

**State of this walkthrough's run:** the backend and frontend were started successfully (backend moved to port **3100** because port 3000 was occupied by another local Next.js app). A local PostgreSQL 18 instance was provisioned, migrated, and seeded with the 10 starter plans; a valid session was used so the authenticated screens render live data. `/health` returns `{"status":"ok"}` , the authenticated API returns all 10 plans, and protected routes return `401` without a session — all as expected.

Local-dev note: `src/core/db/client.js` was given a one-line tweak to skip SSL for `localhost` database URLs (remote Supabase still uses SSL). This is uncommitted and only enables local Postgres.

### Current Gaps / Notes

- `actionsController.updateOpportunity` and one `getMappers` handler are placeholder stubs.
- Vitest is configured but no test files exist yet.
- README is a single line; this document is intended to fill that gap.

— End of overview. Prepared for CSM Synergy from the `CSMSYNERGY/buildbridge` source.